



1 - PURPOSE

Solid Lipid Nanoparticles (SLN) and Nanostructured Lipid Carriers (NLC) are a new generation of colloidal systems, developed during the last decades as nanoparticulate delivery systems for poorly water-soluble drugs. These formulations are fine lipidic emulsions with droplets size of about 50-1000 nm, in which the liquid lipid of a regular emulsion is replaced by a solid core [1]. This core is composed of triglycerides, glyceride mixtures or waxes that are solid at both room and human body temperature. One limiting factor for SLN is that the drug could be expelled from the carrier due to the polymorphism of the solid lipid, hence reducing the drug loading capacity. This problem can be overcome by NLC, in which the presence of a liquid lipid gives more flexibility to the lipid core than in SLN. It results in a reduction of the drug expulsion during solid-to-solid phase transitions. As many other drug delivery systems aimed to increase the bioavailability of poorly water-soluble drugs, these nanoparticulate delivery systems contain lipids and surfactants. The molecules generally used are triacylglycerols (TAGs), diacylglycerols (DAGs), monoacylglycerols (MAGs), phospholipids, polysorbates, and polyethylene-glycols (PEGs) mono- and diesters. All these molecules hold ester bonds that can be cleaved by lipolytic enzymes [2]. Therefore it is necessary to perform *in vitro* lipid digestion to evaluate the fate of lipid-based formulations after oral administration. The aim of this study was to investigate the digestibility of SLN and NLC composed of long chain saturated fatty acid esters (glyceryl distearate, Precirol® ATO 5).

2 - METHODS

• Preparation of nanoparticulate systems

SLN and NLC are prepared by high shear homogenization and ultrasonication technique. Briefly, the aqueous phase is prepared dissolving 240 mg of polysorbates (Tween® 80) and 240 mg of poloxamers 407 (Pluronic® F127) in 60 mL of MilliQ water, heated at the melting temperature of the lipidic phase. On the other hand, the lipidic phase is composed by 480 mg of glyceryl distearate (Precirol® ATO 5) for SLN and of 288 mg of Precirol® ATO 5 and 192 mg of medium chain triglycerides (Labrafac™ Lipophile) for NLC. The lipidic phase is melted at 60°C and, after the addition of the aqueous phase, the mixture is homogenized at 24 000 rpm for 5 min (Ultraturax, IKA). After homogenization, the nanodispersion is rapidly cooled down at -20°C for 10 minutes, to obtain the solidification of the lipidic droplets produced in the emulsification step. Finally, SLN and NLC are sonicated for 15 minutes to reduce nanoparticle aggregation and then stored at +4°C until use.

• Particle size analysis

The dimensional characterization of SLN and NLC is performed by Photon Correlation Spectroscopy (NICOMP Particle Size Systems, USA). Nanoparticles are diluted in MilliQ water and filtered through a 1 µm-syringe filter (Acrodisc® 25 mm syringe filter, 1 µm glass fiber membrane) before analysis.

• In vitro digestion

The experimental setup consists in a pH stat apparatus (Metrohm® AG, Switzerland), comprising a Titrando 802 propeller stirrer/804 Ti Stand combination, a glass pH electrode (iUnitrode) and one 800 Dosino dosing units coupled to 5 mL autoburette. The apparatus is connected to a PC and operated using Tiamo 2.0 software.

The digestion buffer is composed by 150 mM NaCl, 4 mM NaTDC, 1.4 mM CaCl₂·2H₂O and 1 mM TRIS, [3]; concentrations are chosen to mimic the typical concentrations in the small intestine [4]. The pH of the lipolysis medium is adjusted with 0.1 N HCl at 5.5 or at 6.25, to simulate fasted or fed conditions, respectively.

The enzymatic suspension is prepared by dissolving porcine pancreatic powder in cold digestion medium (+4°C), at a concentration of 0.2 mg/mL. The pancreatic suspension is prepared fresh before each test. It is stirred for 15 minutes and then centrifuged at 4000 rpm (2800 g), +4°C for 15 minutes (Universal centrifuge 320R) and the supernatant is collected [5]. The volume of the enzymatic suspension used in the lipolysis tests depends on the fasted or fed conditions, respectively 4 or 22 mL.

For each experiment, 4 mL of SLN or NLC are introduced in the reaction vessel containing 30 mL of lipolysis buffer at 37°C. Continuous mixing is achieved using an overhead propeller stirrer 25 mm in diameter and rotating at approximately 450 rpm. Digestion is started by adding the pancreatic enzyme (t=0 min) and is monitored until 1 hour. Sodium hydroxide is automatically added to the reaction vessel, to maintain a constant pH during digestion. To follow the degradation of lipid nanoparticles, aliquots are sampled at: t=0, t=5, t=15, t=30, t=45 and t=60 min. To stop the digestion 500 µL samples are immediately mixed with 1 mL of acetonitrile and kept at +4°C until gas chromatography analysis (Agilent Gas Chromatograph 7890 A). Three microliters of each sample are injected on a Thermo TG-5MT column (15m x 250 µm x 0.1 µm) with a split ratio of 10:1. The mobile phase is helium, heated at 350°C, with a flow of 1 mL/min. Analytes are detected by a flame ionization detector (FID).

3 - RESULTS AND DISCUSSION

• Particle size analysis

No significant difference was observed either in term of particle size or polydispersity index after dispersion in lipolysis medium, at both pH, and in MilliQ water (Figure 1). All nanoparticles (NP) present a mean diameter between 150 and 200 nm, and a polydispersity index of 0.5.

• In vitro digestion

4-bromophenylboronic acid is classically used as lipase inhibitor during pH-stat analysis [5], but was not adapted to our experimental setup because it reacted with the free hydroxyl group of MAG and DAG to form new chemical entities. Hence, these glycerides could not be quantified by GC analysis. Therefore, a new method was used, consisting of an enzymatic inhibition via acetonitrile addition. The organic solvent, water-miscible, is able to inhibit the lipases causing their precipitation.

For SLN, both in fasted (Figure 2) and fed (data not shown) state conditions, their degradation was completed in less than 5 minutes. Neither at pH 5.5, nor at pH 6.25, MAGs were anymore detectable, even at t=0 (before the addition of the enzyme). We hypothesize that surfactants have micellized MAGs, which subsequently were not extracted by the organic phase.

To assess if the degradation of TAGs and DAGs was caused by the action of the enzyme and not by the autolysis of the NP, the same experiments were repeated without enzymes. No degradation of NP was observed in the fasted (Figure 2) and fed (data not shown) state conditions without enzyme. The profile of NLC digestion in fasted conditions is shown in Figure 3. For TAGs and DAGs the digestion pattern is very similar to the one obtained for SLN. In this case, MAGs could also be assayed by GC-FID. We can hypothesize that the presence of Labrafac™ Lipophile changed the polarity of the sample resulting in a good and complete extraction of lipids from the digested samples. At pH 5.5 the digestion of MAGs seems slightly slower than pH 6.25, with a reduction of only about 22% of MAGs in the first 5 minute of lipolysis, and 100% of digestion after 10 additional minutes. In the fed state conditions (pH 6.25), all glycerides are digested in less than 5 minutes due to the higher quantity of enzymatic suspension added to the reaction vessel (22 mL in fed instead of 4 mL in fasted condition, data not shown).

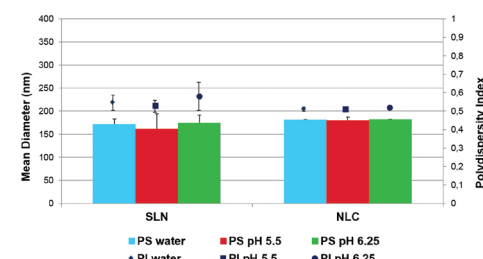


Figure 1. Particle Size (PS) and Polydispersity Index (PI) of SLN and NLC in water, lipolysis media pH 5.5 and pH 6.25 (mean ± s.d., n=3).

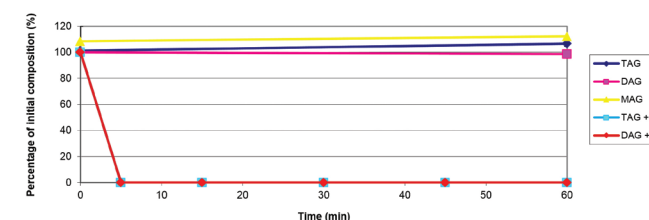


Figure 2. Evolution of the composition of Precirol® ATO 5 in SLN during in vitro lipolysis at pH 5.5
TAG: triacylglycerols, DAG: diacylglycerols, MAG: monoacylglycerols, E: experiments performed with enzymes.

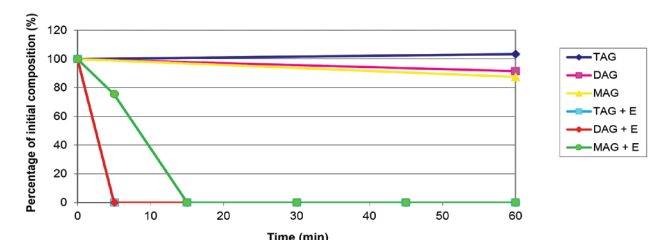


Figure 3. Evolution of the composition of Precirol® ATO 5 in NLC during in vitro lipolysis at pH 5.5
TAG: triacylglycerols, DAG: diacylglycerols, MAG: monoacylglycerols, E: experiments performed with enzymes.

4 - CONCLUSIONS

SLN and NLC systems obtained with long-chain saturated glycerides are rapidly and completely digested by pancreatic enzymes. pH-stat titration appears to be a powerful technique to follow the digestibility of these solid lipid-based nanoparticles.

5 – REFERENCES

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